

# Package ‘broadcast’

June 21, 2025

**Title** 'Numpy'-

Like Broadcasted Operations for Atomic and Recursive Arrays with Minimal Dependencies

**Version** 0.0.0.9008

**Description** Implements efficient broadcasted operations for atomic and recursive arrays.

Besides linking to 'Rcpp',

'broadcast' does not depend on, vendor, link to, include, or otherwise use any external libraries;

'broadcast' was essentially made from scratch and can be installed out-of-the-box.

The implementations available in 'broadcast' include, but are not limited to, the following.

- 1) Broadcasted element-wise operations on any 2 arrays;  
they support a large set of relational-, arithmetic-, Boolean-, string- and bit-wise operations.
- 2) A faster, more memory efficient, and broadcasted abind-like function,  
for binding arrays along an arbitrary dimension.
- 3) Broadcasted ifelse- and apply-like functions.
- 4) The `acast()` function, for casting/pivoting an array into a new dimension.

The functions in the 'broadcast' package strive to minimize computation time and memory usage  
(which is not just good for efficient computing, but also for the environment).

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**Encoding** UTF-8

**LinkingTo** Rcpp

**Roxygen** list(markdown = TRUE)

**RoxygenNote** 7.3.2

**Depends** R (>= 4.2.0)

**Imports** Rcpp (>= 1.0.14),  
methods

**Suggests** tinytest, abind, roxygen2

**URL** <https://github.com/tony-aw/broadcast>, <https://tony-aw.github.io/broadcast/>

**BugReports** <https://github.com/tony-aw/broadcast/issues/>

**Language** en-gb

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aaa00\_broadcast\_help    *broadcast Package Overview*

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## Description

broadcast:  
'Numpy'-Like Broadcasted Operations for Atomic and Recursive Arrays with Minimal Dependencies.

Implements efficient broadcasted operations for atomic and recursive arrays.

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The implementations available in 'broadcast' include, but are not limited to, the following:

1. Broadcasted element-wise operations on any 2 arrays; they support a large set of relational-, arithmetic-, Boolean-, string- and bit-wise operations.
2. A faster, more memory efficient, and broadcasted abind-like function, for binding arrays along an arbitrary dimension.
3. Broadcasted ifelse- and apply-like functions.
4. The acast() function, for casting/pivoting an array into a new dimension.

The functions in the 'broadcast' package strive to minimize computation time and memory usage (which is not just good for efficient computing, but also for the environment).

## Links to Get Started

- GitHub main page: <https://github.com/tony-aw/broadcast>
- Quick Start Guide: [https://tony-aw.github.io/broadcast/vignettes/b\\_quickstart.html](https://tony-aw.github.io/broadcast/vignettes/b_quickstart.html)
- Reporting Issues: <https://github.com/tony-aw/broadcast/issues>

## Functions

### Functions for broadcasted element-wise binary operations

'broadcast' provides a set of functions for broadcasted element-wise binary operations with broadcasting.

These functions use an API similar to the [outer](#) function.

The following functions for binary operations are available:

- [bc.b](#): Boolean (i.e. logical) operations;
- [bc.i](#): integer arithmetic and relational operations;
- [bc.d](#): decimal arithmetic and relational operations;
- [bc.cplx](#): complex arithmetic and (in)equality operations;
- [bc.str](#): string (in)equality, concatenation, and distance operations;
- [bc.raw](#): byte- and relational operations for vectors/arrays of type raw;
- [bc.bit](#): BIT-WISE operations, supporting the raw and integer types;
- [bc.list](#): apply any 'R' function to 2 recursive arrays with broadcasting.

`bind_array()`

'broadcast' provides the [bind\\_array](#) function, to bind arrays along an arbitrary dimension, with support for broadcasting.

The API of `bind_array()` is inspired by the fantastic `abind::abind()` function by Tony Plare & Richard Heiberger (2016).

But `bind_array()` differs considerably from `abind::abind` in the following ways:

- `bind_array()` differs from `abind::abind` in that it can handle recursive arrays properly (the `abind::abind` function would unlist everything to atomic arrays, ruining the structure).
- `bind_array()` allows for broadcasting, while `abind::abind` does not support broadcasting.
- `bind_array()` is generally faster than `abind::abind`, as `bind_array()` relies heavily on 'C' and 'C++' code.
- unlike `abind::abind`, `bind_array()` only binds (atomic/recursive) arrays and matrices. `bind_array()` does not attempt to convert things to arrays when they are not arrays, but will give an error instead.  
This saves computation time and prevents unexpected results.
- `bind_array()` has more streamlined naming options, compared to `abind::abind`.

### General functions

'broadcast' also comes with 2 general broadcasted functions:

- [bc.ifelse](#): Broadcasted version of [ifelse](#).
- [bc.apply](#): Broadcasted apply-like function.

### Other functions

'broadcast' provides the [acast](#) function, for casting (i.e. pivoting) an array into a new dimension.

'broadcast' also provides [type-casting](#) functions, which preserve names and dimensions - convenient for arrays.

### Overloading

Sometimes broadcasting is needed in large mathematical expression, involving multiple variables, where precedence is of importance.

For example in an expression like  $x + y / z^y$ .

For such cases, you may want to overload the base operators.

To that end, the 'broadcast' package provides the [broadcaster](#) class, which comes with its own method dispatch for the base operators.

### Author(s)

**Author, Maintainer:** Tony Wilkes <tony\_a\_wilkes@outlook.com> ([ORCID](#))

### References

Plate T, Heiberger R (2016). *abind: Combine Multidimensional Arrays*. R package version 1.4-5, <https://CRAN.R-project.org/package=abind>.

Harris, C.R., Millman, K.J., van der Walt, S.J. et al. *Array programming with NumPy*. Nature 585, 357–362 (2020). doi:10.1038/s4158602026492. ([Publisher link](#)).

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aaa01\_broadcast\_attributes

*Attribute handling*

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### Description

The `bc.` functions and the overloaded operators generally do **not** preserve attributes, unlike the base 'R' operators.

Broadcasting often results in an object with more dimensions, larger dimensions, and/or larger length than the original objects.

Therefore, the names, dimnames, and dim attributes often no longer fit the new object.

Moreover, some classes are only appropriate for certain dimensions or lengths.

The implicit `matrix` class, for example, presumes an object to have exactly 2 dimensions.

And the various classes provided by the 'bit' package have length-related attributes.

So even class attributes cannot be guaranteed to hold for the resulting objects.

However, the `bc.` functions and the overloaded operators **always** preserve the "broadcaster" attribute, as this is necessary to chain together broadcasted operations.

Notice that this contrasts with base 'R' in the following sense:

In base 'R', logical (&, |) and relational (==, !=, etc.) operators never preserve attributes, whereas the broadcasted equivalents do preserve the "broadcaster" attribute.

Almost all functions provided by 'broadcast' are S3 or S4 generics; methods can be written for them for specific classes, so that class-specific attributes can be supported when needed.

Unary operations (i.e.  $+x$ ,  $-x$ ) return the original object, with only the sign adjusted.

---

acast

*Simple and Fast Casting/Pivoting of an Array*


---

## Description

The `acast()` function spreads subsets of an array margin over a new dimension. Written in 'C' and 'C++' for high speed and memory efficiency.

Roughly speaking, `acast()` can be thought of as the "array" analogy to `data.table::dcast()`. But note 2 important differences:

- `acast()` works on arrays instead of `data.tables`.
- `acast()` casts into a completely new dimension (namely `ndim(x) + 1`), instead of casting into new columns.

## Usage

```
acast(x, ...)

## Default S3 method:
acast(
  x,
  margin,
  grp,
  fill = FALSE,
  fill_val = if (is.atomic(x)) NA else list(NULL),
  ...
)
```

## Arguments

|                     |                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>      | an atomic or recursive array.                                                                                                                                                                                                                                                                                                                                                                  |
| <code>...</code>    | further arguments passed to or from methods.                                                                                                                                                                                                                                                                                                                                                   |
| <code>margin</code> | a scalar integer, specifying the margin to cast from.                                                                                                                                                                                                                                                                                                                                          |
| <code>grp</code>    | a factor, where <code>length(grp) == dim(x)[margin]</code> , with at least 2 unique values, specifying which indices of <code>dim(x)[margin]</code> belong to which group. Each group will be cast onto a separate index of dimension <code>ndim(x) + 1</code> . Unused levels of <code>grp</code> will be dropped. Any NA values or levels found in <code>grp</code> will result in an error. |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fill</code>     | <p>Boolean.</p> <p>When factor <code>grp</code> is unbalanced (i.e. has unequally sized groups) the result will be an array where some slices have missing values, which need to be filled. If <code>fill = TRUE</code>, an unbalanced <code>grp</code> factor is allowed, and missing values will be filled with <code>fill_val</code>.</p> <p>If <code>fill = FALSE</code> (default), an unbalanced <code>grp</code> factor is not allowed, and providing an unbalanced factor for <code>grp</code> produces an error.</p> <p>When <code>x</code> has type of <code>raw</code>, unbalanced <code>grp</code> is never allowed.</p> |
| <code>fill_val</code> | <p>scalar of the same type of <code>x</code>, giving value to use to fill in the gaps when <code>fill = TRUE</code>.</p> <p>The <code>fill_val</code> argument is ignored when <code>fill = FALSE</code> or when <code>x</code> has type of <code>raw</code>.</p>                                                                                                                                                                                                                                                                                                                                                                   |

## Details

For the sake of illustration, consider a matrix `x` and a grouping factor `grp`. Let the integer scalar `k` represent a group in `grp`, such that  $k \in 1:nlevels(grp)$ . Then the code

```
out <- acast(x, margin = 1, grp = grp)
```

essentially performs the following for every group `k`:

- copy-paste the subset `x[grp == k, ]` to the subset `out[, , k]`.

Please see the examples section to get a good idea on how this function casts an array. A more detailed explanation of the `acast()` function can be found on the website.

## Value

An array with the following properties:

- the number of dimensions of the output array is equal to `ndim(x) + 1`;
- the dimensions of the output array is equal to `c(dim(x), max(tabulate(grp)))`;
- the `dimnames` of the output array is equal to `c(dimnames(x), list(levels(grp)))`.

## Back transformation

From the casted array,

```
out = acast(x, margin, grp),
```

one can get the original `x` back by using

```
back = asplit(out, ndim(out)) |> bind_array(along = margin).
```

Note, however, the following about the back-transformed array `back`:

- `back` will be ordered by `grp` along dimension `margin`;
- if the levels of `grp` did not have equal frequencies, then `dim(back)[margin] > dim(x)[margin]`, and `back` will have more missing values than `x`.

## Examples

```
x <- cbind(id = c(rep(1:3, each = 2), 1), grp = c(rep(1:2, 3), 2), val = rnorm(7))
print(x)

grp <- as.factor(x[, 2])
levels(grp) <- c("a", "b")
margin <- 1L

acast(x, margin, grp, fill = TRUE)
```

bc.b

*Broadcasted Boolean Operations*

## Description

The `bc.b()` function performs broadcasted Boolean operations on 2 arrays.

Please note that these operations will treat the input as logical.

Therefore, something like `bc.b(1, 2, "==")` returns TRUE, because both 1 and 2 are TRUE when treated as logical.

## Usage

```
bc.b(x, y, op, ...)
```

## S4 method for signature 'ANY'

```
bc.b(x, y, op)
```

## Arguments

|                   |                                                                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x, y</code> | conformable arrays of type logical, integer(32 bit), or raw.                                                                                                                                                                                                             |
| <code>op</code>   | a single string, giving the operator.<br>Supported Boolean operators: <code>&amp;</code> , <code> </code> , <code>xor</code> , <code>nand</code> , <code>==</code> , <code>!=</code> , <code>&lt;</code> , <code>&gt;</code> , <code>&lt;=</code> , <code>&gt;=</code> . |
| <code>...</code>  | further arguments passed to or from methods.                                                                                                                                                                                                                             |

## Details

`bc.b()` efficiently casts the input to logical without making copies of the entire vectors/arrays. Since the input is treated as logical, the following equalities hold for `bc.b()`:

- `"=="` is equivalent to `(x & y) | (!x & !y)`, but faster;
- `"!="` is equivalent to `xor(x, y)`;
- `"<"` is equivalent to `(!x & y)`, but faster;
- `">"` is equivalent to `(x & !y)`, but faster;
- `"<="` is equivalent to `(!x & y) | (y == x)`, but faster;
- `">="` is equivalent to `(x & !y) | (y == x)`, but faster.

**Value**

A logical array as a result of the broadcasted Boolean operation.

**Examples**

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(TRUE, FALSE, NA), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))

bc.b(x, y, "&")
bc.b(x, y, "|")
bc.b(x, y, "xor")
bc.b(x, y, "nand")
bc.b(x, y, "==")
bc.b(x, y, "!=")
```

---

bc.bit

---

*Broadcasted Bit-wise Operations*


---

**Description**

The `bc.bit()` function performs broadcasted bit-wise operations on pairs of arrays, where both arrays are of type raw or both arrays are of type integer.

**Usage**

```
bc.bit(x, y, op, ...)

## S4 method for signature 'ANY'
bc.bit(x, y, op)
```

**Arguments**

|                   |                                                                                                                                                                                                                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x, y</code> | conformable raw or integer (32 bit) vectors or arrays.                                                                                                                                                                                                                                                                    |
| <code>op</code>   | a single string, giving the operator.<br>Supported bit-wise operators: <code>&amp;</code> , <code> </code> , <code>xor</code> , <code>nand</code> , <code>==</code> , <code>!=</code> , <code>&lt;</code> , <code>&gt;</code> , <code>&lt;=</code> , <code>&gt;=</code> , <code>&lt;&lt;</code> , <code>&gt;&gt;</code> . |
| <code>...</code>  | further arguments passed to or from methods.                                                                                                                                                                                                                                                                              |



## Details

The "&", "|", "xor", and "nand" operators given in `bc.bit()` perform BIT-WISE AND, OR, XOR, and NAND operations, respectively.

The relational operators given in `bc.bit()` perform BIT-WISE relational operations:

- "==" is equivalent to bit-wise  $(x \& y) \mid (!x \& !y)$ , but faster;
- "!=" is equivalent to bit-wise  $xor(x, y)$ ;
- "<" is equivalent to bit-wise  $(!x \& y)$ , but faster;
- ">" is equivalent to bit-wise  $(x \& !y)$ , but faster;
- "<=" is equivalent to bit-wise  $(!x \& y) \mid (y == x)$ , but faster;
- ">=" is equivalent to bit-wise  $(x \& !y) \mid (y == x)$ , but faster.

The "<<" and ">>" operators perform bit-wise left-shift and right-shift, respectively, on  $x$  by unit  $y$ . For these shift operations,  $y$  being larger than the number of bits of  $x$  results in an error. Shift operations are only supported for type of raw.

## Value

For bit-wise operators:

An array of the same type as  $x$  and  $y$ , as a result of the broadcasted bit-wise operation.

## Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- as.raw(0:10)
y.data <- as.raw(10:0)
x <- array(x.data, x.dim)
y <- array(y.data, c(4,1,1))

bc.bit(x, y, "&")
bc.bit(x, y, "|")
bc.bit(x, y, "xor")
```

## Description

The `bc.cplx()` function performs broadcasted complex numeric operations on pairs of arrays.

Note that `bc.cplx()` uses more strict NA checks than base 'R':

If for an element of either  $x$  or  $y$ , either the real or imaginary part is NA or NaN, then the result of the operation for that element is necessarily NA.

**Usage**

```
bc.cplx(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
```

```
bc.cplx(x, y, op)
```

**Arguments**

|                   |                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <code>x, y</code> | conformable atomic arrays of type complex.                                                                                      |
| <code>op</code>   | a single string, giving the operator.<br>Supported arithmetic operators: +, -, *, /.<br>Supported relational operators: ==, !=. |
| <code>...</code>  | further arguments passed to or from methods.                                                                                    |

**Value**

For arithmetic operators:  
A complex array as a result of the broadcasted arithmetic operation.

For relational operators:  
A logical array as a result of the broadcasted relational comparison.

**Examples**

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
gen <- function() sample(c(rnorm(10), NA, NA, NaN, NaN, Inf, Inf, -Inf, -Inf))
x <- array(gen() + gen() * -1i, x.dim)
y <- array(gen() + gen() * -1i, c(4,1,1))

bc.cplx(x, y, "==")
bc.cplx(x, y, "!=")

bc.cplx(x, y, "+")

bc.cplx(array(gen() + gen() * -1i), array(gen() + gen() * -1i), "==")
bc.cplx(array(gen() + gen() * -1i), array(gen() + gen() * -1i), "!=")

x <- gen() + gen() * -1i
y <- gen() + gen() * -1i
out <- bc.cplx(array(x), array(y), "*")
cbind(x, y, x*y, out)
```

bc.d

*Broadcasted Decimal Numeric Operations***Description**

The `bc.d()` function performs broadcasted decimal numeric operations on 2 numeric or logical arrays.

**Usage**

```
bc.d(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
```

```
bc.d(x, y, op, tol = sqrt(.Machine$double.eps))
```

**Arguments**

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x, y</code> | conformable logical or numeric arrays.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>op</code>   | a single string, giving the operator.<br>Supported arithmetic operators: <code>+</code> , <code>-</code> , <code>*</code> , <code>/</code> , <code>^</code> , <code>pmin</code> , <code>pmax</code> .<br>Supported relational operators: <code>==</code> , <code>!=</code> , <code>&lt;</code> , <code>&gt;</code> , <code>&lt;=</code> , <code>&gt;=</code> , <code>d==</code> , <code>d!=</code> , <code>d&lt;</code> , <code>d&gt;</code> , <code>d&lt;=</code> , <code>d&gt;=</code> . |
| <code>...</code>  | further arguments passed to or from methods.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>tol</code>  | a single number between 0 and 0.1, giving the machine tolerance to use.<br>Only relevant for the following operators:<br><code>d==</code> , <code>d!=</code> , <code>d&lt;</code> , <code>d&gt;</code> , <code>d&lt;=</code> , <code>d&gt;=</code><br>See the <code>%d==%</code> , <code>%d!=%</code> , <code>%d&lt;%</code> , <code>%d&gt;%</code> , <code>%d&lt;=%</code> , <code>%d&gt;=%</code> operators from the 'tinycodet' package for details.                                    |

**Value**

For arithmetic operators:

A numeric array as a result of the broadcasted decimal arithmetic operation.

For relational operators:

A logical array as a result of the broadcasted decimal relational comparison.

**Examples**

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(NA, 1.1:1000.1), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))
```

```
bc.d(x, y, "+")
bc.d(x, y, "-")
bc.d(x, y, "*")
bc.d(x, y, "/")
bc.d(x, y, "^")
```

```
bc.d(x, y, "==")
bc.d(x, y, "!=")
bc.d(x, y, "<")
bc.d(x, y, ">")
bc.d(x, y, "<=")
bc.d(x, y, ">=")
```

---

|      |                                                                              |
|------|------------------------------------------------------------------------------|
| bc.i | <i>Broadcasted Integer Numeric Operations with Extra Overflow Protection</i> |
|------|------------------------------------------------------------------------------|

---

## Description

The `bc.i()` function performs broadcasted integer numeric operations on 2 numeric or logical arrays.

Please note that these operations will treat the input as (double typed) integers, and will efficiently truncate when necessary.

Therefore, something like `bc.i(1, 1.5, "==")` returns TRUE, because `trunc(1.5)` equals 1.

## Usage

```
bc.i(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
bc.i(x, y, op)
```

## Arguments

|      |                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x, y | conformable logical or numeric arrays.                                                                                                                                                                                                                                        |
| op   | a single string, giving the operator.<br>Supported arithmetic operators: +, -, *, gcd, %%, %/%, ^, pmin, pmax.<br>Supported relational operators: ==, !=, <, >, <=, >=.<br>The "gcd" operator performs the Greatest Common Divisor" operation, using the Euclidean algorithm. |
| ...  | further arguments passed to or from methods.                                                                                                                                                                                                                                  |

## Value

For arithmetic operators:

A numeric array of whole numbers, as a result of the broadcasted arithmetic operation.

Base 'R' supports integers from  $-2^{53}$  to  $2^{53}$ , which thus range from approximately -9 quadrillion

to +9 quadrillion.

Values outside of this range will be returned as -Inf or Inf, as an extra protection against integer overflow.

For relational operators:

A logical array as a result of the broadcasted integer relational comparison.

### Examples

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(NA, 1.1:1000.1), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))
```

```
bc.i(x, y, "+")
bc.i(x, y, "-")
bc.i(x, y, "*")
bc.i(x, y, "gcd") # greatest common divisor
bc.i(x, y, "^")
```

```
bc.i(x, y, "==")
bc.i(x, y, "!=")
bc.i(x, y, "<")
bc.i(x, y, ">")
bc.i(x, y, "<=")
bc.i(x, y, ">=")
```

---

bc.list

*Broadcasted Operations for Recursive Arrays*

---

### Description

The bc.list() function performs broadcasted operations on 2 Recursive arrays.

### Usage

```
bc.list(x, y, f, ...)
```

```
## S4 method for signature 'ANY'
bc.list(x, y, f)
```

### Arguments

|      |                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| x, y | conformable Recursive arrays (i.e. arrays of type list).                                                                           |
| f    | a function that takes in exactly <b>2</b> arguments, and <b>returns</b> a result that can be stored in a single element of a list. |
| ...  | further arguments passed to or from methods.                                                                                       |

**Value**

A recursive array.

**Examples**

```
x.dim <- c(10, 2,2)
x.len <- prod(x.dim)

gen <- function(n) sample(list(letters, month.abb, 1:10), n, TRUE)

x <- array(gen(10), x.dim)
y <- array(gen(10), c(10,1,1))

bc.list(
  x, y,
  \(x, y) c(length(x) == length(y), typeof(x) == typeof(y))
)
```

---

bc.raw

*Broadcasted Byte and Relational Operations on Raw Arrays*


---

**Description**

The `bc.raw()` function performs broadcasted byte- and relational operations on arrays of type `raw`.

For bit-wise operations, use [bc.bit](#).

For logical operations, use [bc.b](#)

**Usage**

```
bc.raw(x, y, op, ...)
```

## S4 method for signature 'ANY'

```
bc.raw(x, y, op)
```

**Arguments**

|                   |                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x, y</code> | conformable raw vectors or arrays.                                                                                                                                                                                                                                                                                                                                       |
| <code>op</code>   | a single string, giving the operator.<br>Supported byte operators: <code>pmin</code> , <code>pmax</code> , <code>diff</code> .<br>The "diff" operator performs the byte equivalent of <code>abs(x - y)</code> .<br>Supported relational operators: <code>==</code> , <code>!=</code> , <code>&lt;</code> , <code>&gt;</code> , <code>&lt;=</code> , <code>&gt;=</code> . |
| <code>...</code>  | further arguments passed to or from methods.                                                                                                                                                                                                                                                                                                                             |

**Value**

For the byte operators:

A array of type raw, as a result of the broadcasted byte operation.

For relational operators:

A logical array as a result of the broadcasted relational comparison.

**Examples**

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- as.raw(0:10)
y.data <- as.raw(10:0)
x <- array(x.data, x.dim)
y <- array(y.data, c(4,1,1))
```

```
bc.raw(x, y, "==")
bc.raw(x, y, "!=")
bc.raw(x, y, "<")
bc.raw(x, y, ">")
bc.raw(x, y, "<=")
bc.raw(x, y, ">=")
```

---

bc.str

*Broadcasted String Operations*


---

**Description**

The `bc.str()` function performs broadcasted string operations on pairs of arrays.

**Usage**

```
bc.str(x, y, op, ...)
```

```
## S4 method for signature 'ANY'
bc.str(x, y, op)
```

**Arguments**

|                   |                                                                                                                                                                                                                                 |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>x, y</code> | conformable atomic arrays of type character.                                                                                                                                                                                    |
| <code>op</code>   | a single string, giving the operator.<br>Supported concatenation operators: <code>+</code> .<br>Supported relational operators: <code>==</code> , <code>!=</code> .<br>Supported distance operators: <code>levenshtein</code> . |
| <code>...</code>  | further arguments passed to or from methods.                                                                                                                                                                                    |

**Value**

For concatenation operation:

A character array as a result of the broadcasted concatenation operation.

For relational operation:

A logical array as a result of the broadcasted relational comparison.

For distance operation:

An integer array as a result of the broadcasted distance measurement.

**References**

The 'C++' code for the Levenshtein edit string distance is based on the code found in [https://rosettacode.org/wiki/Levenshtein\\_distance#C++](https://rosettacode.org/wiki/Levenshtein_distance#C++)

**Examples**

```
# string concatenation:
x <- array(letters, c(10, 2, 1))
y <- array(letters, c(10,1,1))
bc.str(x, y, "+")

# string (in)equality:
bc.str(array(letters), array(letters), "==")
bc.str(array(letters), array(letters), "!=")

# string distance (Levenshtein):
x <- array(month.name, c(12, 1))
y <- array(month.abb, c(1, 12))
out <- bc.str(x, y, "levenshtein")
dimnames(out) <- list(month.name, month.abb)
print(out)
```

---

bcapply

---

*Apply a Function to 2 Broadcasted Arrays*


---

**Description**

The `bcapply()` function applies a function to 2 arrays element-wise with broadcasting.

**Usage**

```
bcapply(x, y, f, ...)
```

## S4 method for signature 'ANY'

```
bcapply(x, y, f, v = NULL)
```



**Arguments**

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x, y | conformable atomic or recursive arrays.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| f    | a function that takes in exactly <b>2</b> arguments, and <b>returns</b> a result that can be stored in a single element of a recursive or atomic array.                                                                                                                                                                                                                                                                                                                                                                      |
| ...  | further arguments passed to or from methods.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| v    | either NULL, or single string, giving the scalar type for a single iteration.<br>If NULL (default) or "list", the result will be a recursive array.<br>If it is certain that, for every iteration, f() always results in a <b>single atomic scalar</b> , the user can specify the type in v to pre-allocate the result.<br>Pre-allocating the results leads to slightly faster and more memory efficient code.<br>NOTE: Incorrectly specifying v leads to undefined behaviour;<br>when unsure, leave v at its default value. |

**Value**

An atomic or recursive array with dimensions bc\_dim(x, y).

**Examples**

```
x.dim <- c(5, 3, 2)
x.len <- prod(x.dim)

gen <- function(n) sample(list(letters, month.abb, 1:10), n, TRUE)

x <- array(gen(10), x.dim)
y <- array(gen(10), c(5, 1, 1))

f <- function(x, y) paste(x, y)
bcapply(x, y, f, v = "character")
```

---

bc\_dim

---

*Predict Broadcasted dimensions*


---

**Description**

bc\_dim(x, y) gives the dimensions an array would have, as the result of an broadcasted binary element-wise operation between 2 arrays x and y.

**Usage**

```
bc_dim(x, y)
```

**Arguments**

|      |                               |
|------|-------------------------------|
| x, y | an atomic or recursive array. |
|------|-------------------------------|

**Value**

Returns an integer vector giving the broadcasted dimension sizes.

**Examples**

```
x.dim <- c(4:2)
x.len <- prod(x.dim)
x.data <- sample(c(TRUE, FALSE, NA), x.len, TRUE)
x <- array(x.data, x.dim)
y <- array(1:50, c(4,1,1))

dim(bc.b(x, y, "&")) == bc_dim(x, y)
dim(bc.b(x, y, "|")) == bc_dim(x, y)
```

---

|           |                           |
|-----------|---------------------------|
| bc_ifelse | <i>Broadcasted Ifelse</i> |
|-----------|---------------------------|

---

**Description**

The `bc_ifelse()` function performs a broadcasted form of `ifelse()`.

**Usage**

```
bc_ifelse(test, yes, no, ...)

## S4 method for signature 'ANY'
bc_ifelse(test, yes, no)
```

**Arguments**

|         |                                                                                                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| test    | a vector or array, with the type logical, integer, or raw, and a length equal to <code>prod(bc_dim(yes, no))</code> .<br>If yes / no are of type raw, test is not allowed to contain any NAs. |
| yes, no | conformable arrays of the same type.<br>All <a href="#">atomic</a> types are supported.<br>Recursive arrays of type <a href="#">list</a> are also supported.                                  |
| ...     | further arguments passed to or from methods.                                                                                                                                                  |

**Value**

The output, here referred to as `out`, will be an array of the same type as `yes` and `no`.  
If `test` has the same dimensions as `bc_dim(yes, no)`, then `out` will also have the same dimnames as `test`.

After broadcasting `yes` against `no`, given any element index `i`, the following will hold for the output:

- when `test[i] == TRUE`, `out[i]` is `yes[i]`;

- when `test[i] == FALSE`, `out[i]` is `no[i]`;
- when `test[i]` is `NA`, `out[i]` is `NA` when `yes` and `no` are atomic, and `out[i]` is `list(NULL)` when `yes` and `no` are recursive.

### Examples

```
x.dim <- c(5, 3, 2)
x.len <- prod(x.dim)

x <- array(sample(1:100), x.dim)
y <- array(sample(1:100), c(5, 1, 1))

cond <- bc.i(x, y, ">")

bc_ifelse(cond, yes = x, no = -1)
```

---

bind\_array

---

*Dimensional Binding of Arrays with Broadcasting*


---

### Description

`bind_array()` binds (atomic/recursive) arrays and (atomic/recursive) matrices.  
Returns an array.  
Allows for broadcasting.

### Usage

```
bind_array(
  input,
  along,
  rev = FALSE,
  ndim2bc = 1L,
  name_along = TRUE,
  comnames_from = 1L
)
```

### Arguments

- |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| input | <p>a list of arrays; both atomic and recursive arrays are supported, and can be mixed.</p> <p>If argument <code>input</code> has length 0, or it contains exclusively objects where one or more dimensions are 0, an error is returned.</p> <p>If <code>input</code> has length 1, <code>bind_array()</code> simply returns <code>input[[1L]]</code>.</p> <p><code>input</code> may not contain more than <math>2^{16}</math> objects.</p>                                                                                                    |
| along | <p>a single integer, indicating the dimension along which to bind the dimensions. I.e. use <code>along = 1</code> for row-binding, <code>along = 2</code> for column-binding, etc.</p> <p>Specifying <code>along = 0</code> will bind the arrays on a new dimension before the first, making <code>along</code> the new first dimension.</p> <p>Specifying <code>along = N + 1</code>, with <code>N = max(1st.ndim(input))</code>, will create an additional dimension (<code>N + 1</code>) and bind the arrays along that new dimension.</p> |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rev           | <p>Boolean, indicating if along should be reversed, counting backwards.</p> <p>If FALSE (default), along works like normally; if TRUE, along is reversed.</p> <p>I.e. along = 0, rev = TRUE is equivalent to along = N+1, rev = FALSE;</p> <p>and along = N+1, rev = TRUE is equivalent to along = 0, rev = FALSE;</p> <p>with N = max(1st.ndim(input)).</p>                                                          |
| ndim2bc       | <p>a single non-negative integer;</p> <p>specify here the maximum number of dimensions that are allowed to be broadcasted when binding arrays.</p> <p>If ndim2bc = 0L, <b>no</b> broadcasting will be allowed at all.</p>                                                                                                                                                                                             |
| name_along    | <p>Boolean, indicating if dimension along should be named.</p> <p>Please run the code in the examples section to get a demonstration of the naming behaviour.</p>                                                                                                                                                                                                                                                     |
| comnames_from | <p>either an integer scalar or NULL.</p> <p>Indicates which object in input should be used for naming the shared dimension.</p> <p>If NULL, no communal names will be given.</p> <p>For example:</p> <p>When binding columns of matrices, the matrices will share the same rownames.</p> <p>Using comnames_from = 10 will then result in bind_array() using rownames(input[[10]]) for the rownames of the output.</p> |

## Value

An array.

## Examples

```
# Simple example ====
x <- array(1:20, c(5, 4))
y <- array(-1:-15, c(5, 3))
z <- array(21:40, c(5, 4))
input <- list(x, y, z)
# column binding:
bind_array(input, 2L)

# Mixing types ====
# here, atomic and recursive arrays are mixed,
# resulting in recursive arrays

# creating the arrays:
x <- c(
  lapply(1:3, \(x)sample(c(TRUE, FALSE, NA))),
  lapply(1:3, \(x)sample(1:10)),
  lapply(1:3, \(x)rnorm(10)),
  lapply(1:3, \(x)sample(letters))
) |> matrix(4, 3, byrow = TRUE)
dimnames(x) <- list(letters[1:4], LETTERS[1:3])
print(x)
```

```

y <- matrix(1:12, 4, 3)
print(y)
z <- matrix(letters[1:12], c(4, 3))

# column-binding:
input <- list(x = x, y = y, z = z)
bind_array(input, along = 2L)

# Illustrating `along` argument ====
# using recursive arrays for clearer visual distinction
input <- list(x = x, y = y)

bind_array(input, along = 0L) # binds on new dimension before first
bind_array(input, along = 1L) # binds on first dimension (i.e. rows)
bind_array(input, along = 2L)
bind_array(input, along = 3L) # bind on new dimension after last

bind_array(input, along = 0L, TRUE) # binds on new dimension after last
bind_array(input, along = 1L, TRUE) # binds on last dimension (i.e. columns)
bind_array(input, along = 2L, TRUE)
bind_array(input, along = 3L, TRUE) # bind on new dimension before first

# binding, with empty arrays ====
emptyarray <- array(numeric(0L), c(0L, 3L))
dimnames(emptyarray) <- list(NULL, paste("empty", 1:3))
print(emptyarray)
input <- list(x = x, y = emptyarray)
bind_array(input, along = 1L, comnames_from = 2L) # row-bind

# Illustrating `name_along` ====
x <- array(1:20, c(5, 3), list(NULL, LETTERS[1:3]))
y <- array(-1:-20, c(5, 3))
z <- array(-1:-20, c(5, 3))

bind_array(list(a = x, b = y, z), 2L)
bind_array(list(x, y, z), 2L)
bind_array(list(a = unname(x), b = y, c = z), 2L)
bind_array(list(x, a = y, b = z), 2L)
input <- list(x, y, z)
names(input) <- c("", NA, "")
bind_array(input, 2L)

```

## Description

`broadcaster()` checks if an array has the "broadcaster" attribute.  
`broadcaster()<-` sets or un-sets the class attribute "broadcaster" on an array.

The broadcaster class attribute exists purely to overload the arithmetic, Boolean, bit-wise, and relational infix operators, to support broadcasting.

This makes mathematical expressions with multiple variables, where precedence may be important, far more convenient.

Like in the following calculation:

$x / (y + z)$

Broadcaster arrays come with their own method dispatch for the base operators.

If at least one of the 2 arguments of the base operators has the broadcaster class attribute, and no other class (like `bit64`) interferes, broadcasting will occur in the same manner as used in the various `bc.*` - functions.

## Usage

```
broadcaster(x)
```

```
broadcaster(x) <- value
```

## Arguments

|                    |                                                                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>x</code>     | object to check or set.<br>Only S3 vectors and arrays are supported, and only up to 16 dimensions.                                      |
| <code>value</code> | set to <code>TRUE</code> to make an array a broadcaster, or <code>FALSE</code> to remove the broadcaster class attribute from an array. |

## Value

For `broadcaster()`:  
`TRUE` if an array or vector is a broadcaster, or `FALSE` if it is not.

For `broadcaster()<-`:  
 Returns nothing, but sets (if right hand side is `TRUE`) or removes (if right hand side is `FALSE`) the "broadcaster" class attribute.

## Examples

```
# maths ====

x <- 1:10
broadcaster(x) <- TRUE

y <- 1:10
broadcaster(y) <- TRUE
```

```

x + y / x
x + 1 # mathematically equivalent to the above, since x == y
(x + y) / x
2 * x/x # mathematically equivalent to the above, since x == y

dim(x) <- c(10, 1)
dim(y) <- c(1, 10)

x + y / x
(x + y) / x

(x + y) * x

# relational operators ====
x <- 1:10
y <- array(1:10, c(1, 10))
broadcaster(x) <- TRUE
broadcaster(y) <- TRUE

x == y
x != y
x < y
x > y
x <= y
x >= y

```

---

ndim

*Get number of dimensions*


---

## Description

ndim() returns the number of dimensions of an object.  
 lst.ndim() returns the number of dimensions of every list-element.

## Usage

```
ndim(x)
```

```
lst.ndim(x)
```

## Arguments

x                      a vector or array (for ndim()), or a list of vectors/arrays (for lst.ndim()).

## Value

For ndim(): an integer scalar.  
 For lst.ndim(): an integer vector, with the same length, names and dimensions as x.

## Examples

```
# matrix example ====
x <- list(
  array(1:10, 10),
  array(1:10, c(2, 5)),
  array(c(letters, NA), c(3,3,3))
)
lst.ndim(x)
```

---

rep\_dim

*Replicate Array Dimensions*

---

## Description

The `rep_dim()` function replicates array dimensions until the specified dimension sizes are reached, and returns the array.

The various broadcasting functions recycle array dimensions virtually, meaning little to no additional memory is needed.

The `rep_dim()` function, however, physically replicates the dimensions of an array (and thus actually occupies additional memory space).

## Usage

```
rep_dim(x, tdim)
```

## Arguments

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| <code>x</code>    | an atomic or recursive array or matrix.                  |
| <code>tdim</code> | an integer vector, giving the target dimension to reach. |

## Value

Returns the replicated array.

## Examples

```
x <- matrix(1:9, 3,3)
colnames(x) <- LETTERS[1:3]
rownames(x) <- letters[1:3]
names(x) <- month.abb[1:9]
print(x)

rep_dim(x, c(3,3,2)) # replicate to larger size
```



## Description

Type casting usually strips away attributes of objects.

The functions provided here preserve dimensions, dimnames, and names, which may be more convenient for arrays and array-like objects.

The functions are as follows:

- `as_bool()`: converts object to atomic type logical (TRUE, FALSE, NA).
- `as_int()`: converts object to atomic type integer.
- `as_dbl()`: converts object to atomic type double (AKA numeric).
- `as_cplx()`: converts object to atomic type complex.
- `as_chr()`: converts object to atomic type character.
- `as_raw()`: converts object to atomic type raw.
- `as_list()`: converts object to recursive type list.

`as_num()` is an alias for `as_dbl()`.

`as_str()` is an alias for `as_chr()`.

See also [typeof](#).

## Usage

```
as_bool(x, ...)
```

```
as_int(x, ...)
```

```
as_dbl(x, ...)
```

```
as_num(x, ...)
```

```
as_chr(x, ...)
```

```
as_str(x, ...)
```

```
as_cplx(x, ...)
```

```
as_raw(x, ...)
```

```
as_list(x, ...)
```

**Arguments**

`x`                    an R object.  
`...`                further arguments passed to or from other methods.

**Value**

The converted object.

**Examples**

```
# matrix example ====
x <- matrix(sample(-1:28), ncol = 5)
colnames(x) <- month.name[1:5]
rownames(x) <- month.abb[1:6]
names(x) <- c(letters[1:20], LETTERS[1:10])
print(x)

as_bool(x)
as_int(x)
as_dbl(x)
as_chr(x)
as_cplx(x)
as_raw(x)

#####

# factor example ====
x <- factor(month.abb, levels = month.abb)
names(x) <- month.name
print(x)

as_bool(as_int(x) > 6)
as_int(x)
as_dbl(x)
as_chr(x)
as_cplx(x)
as_raw(x)
```

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